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Symbolically,

4K + 0L= ₹.120

3K + 3L= ₹.120

2K + 6L= ₹.120

1K + 9L= ₹.120, and

0K + 12L= ₹.120.

Thus, all the combinations A, B, C, D and E cost the same total expenditure.

From the figure 3.10, it is shown that the costs to be incurred on capital and labour are represented by the triangle OAE. The line AE is called as Iso-cost line.

3.10 Producer's Equilibrium

Producer equilibrium implies the situation where producer maximizes his output. It is also known as *optimum combination of the factors of production*. In short, the producer manufactures a given amount of output with *'least cost combination of factors'*, with his given budget.

Optimum Combination of Factors implies either

- there is output maximisation for given inputs or
- there is cost minimisation for the given output.

Conditions for Producer Equilibrium

The two conditions that are to be fulfilled for the attainment of producer equilibrium are:

- The iso-cost line must be tangent to iso-quant curve.
- At point of tangency, the iso-quant curve must be convex to the origin or $MRTS_{Lk}$ must be declining.

When the outlay and prices of two factors, namely, labour and capital are given, producers attain equilibrium (or least cost combination of factors is attained by the firm) where the iso-cost line is tangent to an iso-product curve. It is illustrated in the following Diagram 3.11.

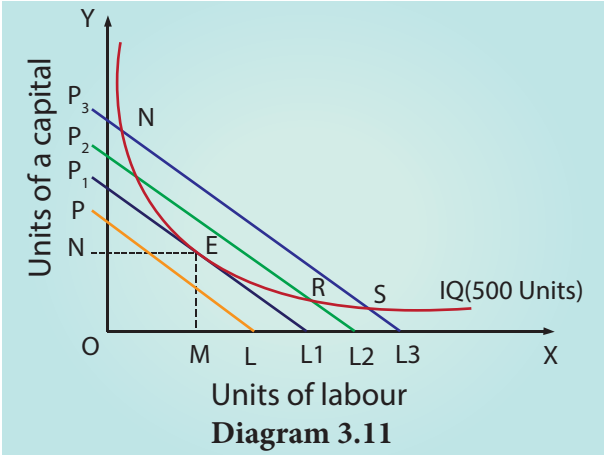


Diagram 3.11

In the above figure, profit of the firm (or the producer) is maximised at the point of equilibrium E.

At the point of equilibrium, the slope of the iso cost line is equal to the slope of iso product curve (or the MRTS of labour for capital is equal to the price ratio of the two factors)

Hence, it can be stated as follows.

$$MRTS_{L,K} = \frac{P_L}{P_K} = 10/30 = 1/3 = 0.333$$

At point E, the firm employs OM units of labour and ON units of capital. In other words, it obtains least cost combination or optimum combination of the two factors to produce the level of output denoted by the iso-quant IQ.